



Border of a String

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Given a string S, find the longest border of the string.

A border is a non-empty substring that is both a prefix and a suffix of the string, but it must be shorter than the entire string. S = "ababa"

Prefixes: "a", "ab", "aba", "abab" Suffixes: "a", "ba", "aba", "baba"

The longest common prefix and suffix is "aba".

Therefore, the answer is:

aba

Input Format

A single line containing the string S.

Constraints

$1 \leq |S| \leq 10^5$ S contains lowercase English letters.

Output Format

Print the longest border of S.

Sample Input 0

ababa

Sample Output 0

aba

Sample Input 1

aaaa

Sample Output 1

aaa

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Language

Python 3



```
1 import sys
2
3 def longest_border():
4     input_data = sys.stdin.read().split()
5     if not input_data:
```

```
5     if not input_data:
6         return
7
8     s = input_data[0]
9     n = len(s)
10
11     if n <= 1:
12         print("")
13         return
14
15
16     lps = [0] * n
17     length = 0
18     i = 1
19
20     while i < n:
21         if s[i] == s[length]:
22             length += 1
23             lps[i] = length
24             i += 1
25         else:
26             if length != 0:
27                 length = lps[length - 1]
28             else:
29                 lps[i] = 0
30             i += 1
```

Line: 43 Col: 1

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Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

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✓ Sample Test case 1

1 | **ababa**

Your Output (stdout)

1 | **aba**

Expected Output

1 | **aba**[Download](#)